Supplementary Information
Dose-agnostic neural deconvolution for low-dose fluorescence microscopy

November 8, 2026

S1. Data, splits, and licenses

Table 1: Public datasets used in this study. BBBC014–018 low-dose measurements are synthesized from the high-SNR references with the forward model of the main text; BBBC039 is used for the downstream segmentation check; the AI4Life pairs are native real noisy/clean acquisitions.

Dataset	Role	FOVs used	Size (pixels)	License
BBBC014 channel 1	train/val/test	56 / 20 / 20	1360×1024	CC BY 3.0
BBBC015	external (5 tune / 15 test)	20 of 144 (channel 0)	1000×768	CC BY 3.0
BBBC016 (GFP)	external (5 tune / 15 test)	20 of 72	512×512	CC BY 3.0
BBBC018 (actin, green)	external (5 tune / 15 test)	20 of 56	512×512	CC BY-NC-SA 3.0
BBBC039 (Hoechst)	segmentation check	100 / 50 / 50	520×696	CC0
AI4Life small pairs	real test	243 held out	256×256	CC BY 4.0
AI4Life large pairs	cross-source	20	1024×1024	CC BY 4.0
FMD fields	second real test	150 of 301 held out	256×256	CC BY 4.0

BBBC014, BBBC015, and BBBC016 are fluorescence images from the Broad Bioimage Benchmark Collection; BBBC015 and BBBC016 are U2OS cytoplasm-to-nucleus assays, and BBBC018 provides a different cell type and actin stain. BBBC015 files encode two channels; only channel 0 is used. The BBBC018 actin signal is in the green channel, which is extracted directly rather than averaged with the blue channel. BBBC039 is used only for the downstream nuclei-segmentation check in S6. The main split is at the field-of-view level; no FOV, well, or acquisition session is shared between training, validation, and test. Test FOVs were never used for early stopping, baseline tuning, target selection, or performance-based figure selection; the qualitative example in Fig. 1 uses a pre-specified test FOV.

The AI4Life-MDC25-Nuclei release (Zenodo 15624741, CC BY 4.0) is the training set of the 2025 AI4Life Microscopy Supervised Denoising Challenge and combines two sources: 2,457 256×256 noisy/clean pairs from the Jacquemet noisy-nuclei dataset and 20 1024×1024 pairs derived from BBBC006. The sources differ in bit depth, saturation, and noise model: the small references reach at most 4095 counts, whereas the large references are clipped at 65535. We split the small pairs by image ID into 1,951/243/243 training/validation/test after excluding 20 IDs that coincide numerically with the large-field IDs as a precaution; the images are not crops of one another (a small image correlates at most 0.33 with any tile of the corresponding large field). Over all 1,951 training images, the maximum test-train thumbnail correlation is 0.55, no pair exceeds 0.8, and the top pair has full-resolution correlation 0.06, so there are no near duplicates (`results/real_leakage_full.json`). The 20 large fields are used only for a cross-source transfer check.

The challenge's hidden test set is not public, so the comparison uses held-out pairs from the public training release, not a leaderboard score.

The AI4Life-MDC25-FMD release (Zenodo 15761900, CC BY 4.0) contains 63,669 256×256 noisy/clean pairs (multiple frames per field) from three microscopes (CF/WF/TP), three channels and four noise levels. We extract one frame from each of 1,998 fields (seed 0) and split by field, stratified by microscope/channel/level, into 1,396/301/301 training/validation/test pairs. The main comparison uses a pre-specified 150-field subset: four fields per microscope/channel/level stratum plus six fifth fields (see `prereg/fmd_subset150.json`).

S2. Extended classical baselines

The main comparison uses TV deconvolution with the known PSF; after joint tuning of weight and ADMM parameters it is the strongest classical method at all three doses. Table 2 reports additional baselines. The NLM [1], Wiener [2], and Richardson–Lucy [3, 4] values use extended parameter grids tuned on 100 validation patches at each dose. Even with the extended grids, none of these methods reaches the network at any dose.

Table 2: Additional classical baselines (PSNR in dB). Doses are $P = 3, 30, 300$ full scale. NLM, Wiener, and Richardson–Lucy use extended parameter grids tuned on 100 validation patches; their values are archived in `results/extended_baselines.json` from a separate run on the same test FOVs. They remain below TV deconvolution and the network.

Method	$P = 3$	$P = 30$	$P = 300$
Noisy input	14.60	24.10	32.48
TV denoising	30.31	34.08	36.26
TV deconvolution	31.34	35.34	38.63
NLM (Anscombe)	28.55	33.31	36.41
Wiener	29.15	33.43	34.79
Richardson–Lucy	29.31	33.62	34.59
U-Net (mixed-dose)	32.27	36.70	40.01

The budget-matched specialists (64 channels, 48k patches, 150 epochs) reach 32.36, 36.72, and 38.68 dB at $P = 3, 30, 100$, within 0.1 dB of the blind model at the same doses (`results/spec_strong.json`). The largest difference is at $P = 3$ (+0.09 dB for the specialist), which is the same order as the patch-draw variation.

S2b. Protocol selection and uncertainty

The primary protocol selects the smallest dose reaching 90% success on the 20 validation FOVs and applies it unchanged to the held-out FOVs (Table 3). Four targets give a 3–3.3× reduction in dose P (equivalently, photon budget) with both methods meeting the test criterion (PSNR 32 and 34 dB, SSIM 0.90, and foreground PSNR 26 dB); three are ties (SSIM 0.85 and 0.92, foreground PSNR 24 dB); and two validation selections miss the test criterion (the U-Net dose at PSNR 30 dB reaches 80%, and the TV dose at foreground PSNR 22 dB reaches 80%), so those factors are not claimed. Table 4 reports the test-oracle doses as an upper bound; because the dose grid has five points and 90% success corresponds to 18/20 FOVs, selection noise is substantial. A FOV bootstrap (2,000 resamples, resampling test FOVs within each dose) gives the oracle

Table 3: Validation-selected fixed doses. The dose is the smallest one reaching 90% success on the 20 validation FOVs and is then applied unchanged to test. A dash under test success means the validation-selected dose did not reach 90% on the held-out FOVs; the factor is then also shown as a dash because it is not claimed.

Metric	Target	TV deconv dose	U-Net dose	Factor	TV test	U-Net test
PSNR	30 dB	10	3	–	100%	–
PSNR	32 dB	30	10	3.0×	100%	100%
PSNR	34 dB	100	30	3.3×	100%	100%
SSIM	0.85	3	3	1.0×	95%	100%
SSIM	0.90	30	10	3.0×	100%	100%
SSIM	0.92	30	30	1.0×	95%	100%
Foreground PSNR	22 dB	3	3	–	–	100%
Foreground PSNR	24 dB	10	10	1.0×	90%	100%
Foreground PSNR	26 dB	100	30	3.3×	100%	100%

Table 4: Test-oracle protocol doses (upper bound): the smallest dose reaching 90% success on the test FOVs. This is not a deployable selection rule and is reported only for comparison with the validation-selected table.

Metric	Target	TV denoise	TV deconv	U-Net
PSNR	30 dB	10	10	10
PSNR	32 dB	100	30	10
PSNR	34 dB	–	100	30
SSIM	0.85	10	3	3
SSIM	0.90	30	30	10
SSIM	0.92	100	30	30
Foreground PSNR	22 dB	10	10	3
Foreground PSNR	24 dB	100	10	10
Foreground PSNR	26 dB	–	100	30

factor distributions in Table 6; these characterize the test-oracle upper bound, not a deployable protocol (`results/protocol_ci.json`). No 10× saving remains after the TV baseline is tuned.

S3. Per-FOV adaptive allocation

An oracle with access to each FOV's full dose-response would save photons (Table 7). A ground-truth-free policy used a cheap preview, ensemble uncertainty, and ridge or k-nearest-neighbour calibration on the 20 validation FOVs. At a 34 dB target, the kNN policy used dose $P = 24.5$ at 100% success versus $P = 30$ for the best fixed dose at 95–96% success, a modest gain. At 35 dB, the best fixed dose at 95% test success was $P = 50$, while the six policies used $P = 52$–77; no policy improved on the fixed protocol. The preview dose is part of the final dose (the final image is acquired by escalating from the preview), so the cost is the final dose, not preview plus final. The gain is therefore target-dependent and not robust enough to replace a fixed dose, which is why the main protocol uses a fixed dose.

Table 5: Ablations (PSNR in dB, matched patches). Top: original base budget, with per-dose specialists (16k patches/60 epochs), the mixed-dose model (24k/100), the dose-conditioned variant (24k/100), the phase-scrambled control (24k/100, conditioned architecture), and tuned TV deconvolution. Middle: the budget-matched specialists, mixed-dose model, dose-conditioned variant, and blind phase-scrambled control (all 64 channels, 48k patches, 150 epochs). Bottom: PSF width at $P = 30$ for the strong mixed-dose model.

Dose	Specialist (under-budget)	Mixed-dose (base)	Conditioned (base)	Scrambled	TV dec
$P = 3$	32.18	32.05	32.20	30.06	31.34
$P = 30$	36.43	36.27	36.44	34.03	35.34
$P = 100$	37.95	38.07	38.29	34.16	37.16

	Specialist (strong)	Mixed-dose (strong)	Conditioned (strong)	Scrambled (strong)	TV dec
$P = 3$	32.36	32.27	32.29	30.19	31.34
$P = 30$	36.72	36.70	36.71	34.46	35.34
$P = 100$	38.68	38.61	38.66	36.18	37.16

σ (pixels)	U-Net	TV deconv
1.5	36.92	35.85
2.0	36.70	35.34
2.5	35.76	34.76

Table 6: Test-oracle protocol doses and FOV-bootstrap factor CIs (2,000 resamples); this is an upper bound, not a validation-selected protocol.

Target	U-Net dose	TV dose	Factor [95% CI]	Bootstrap ≥ threshold
PSNR 30 dB	10	10	1.0 [1, 3.3]	20.9% ≥ 3×
PSNR 32 dB	10	30	3.0 [1, 3]	96.6% ≥ 3×
PSNR 34 dB	30	100	3.3 [3.3, 3.3]	99.9% ≥ 3×
SSIM 0.85	3	3	1.0 [1, 3.3]	7.9% ≥ 3×
SSIM 0.90	10	30	3.0 [1, 3]	81.3% ≥ 3×
SSIM 0.92	30	30	1.0 [1, 3.3]	48.4% ≥ 3×
FG PSNR 22 dB	3	10	3.3 [1, 3.3]	79.5% ≥ 3×
FG PSNR 24 dB	10	10	1.0 [1, 3]	32.8% ≥ 3×
FG PSNR 26 dB	30	100	3.3 [1, 3.3]	97.0% ≥ 3×

S4. Uncertainty details

The ensemble contains three dose-conditioned networks trained with seeds 0, 1, and 2 with the original 32-channel/24k-patch budget. That base-budget conditioned model is 0.07–0.32 dB below the strong mixed-dose main model; the ensemble is used as a proxy for uncertainty behavior, and its absolute calibration against the stronger model is not tested. The uncertainty score is the mean per-pixel standard deviation across the three predictions. At $P = 3$ and 10, 15 and 11 of the 20 test FOVs, respectively, fall below 34 dB; the corresponding AUCs are 0.96 (95% CI 0.84–1.00) and 0.96 (0.85–1.00; `results/uncertainty_ci.json`). At $P = 30$ only one FOV falls below the target, so the AUC is not reported. The uncertainty also correlates with reconstruction RMSE (Spearman $\rho = 0.92$ at $P = 3$). The uncertainty maps are not a proof of artifact-free reconstruction; they identify fields where the ensemble disagrees and where additional light is most useful.

Table 7: Adaptive allocation at two targets. "Fixed" is the dose selected on validation for 90% success; "best fixed at test" is the smallest fixed dose that matches the reported test success; "oracle" uses each FOV's own curve. Doses are P in full-scale units.

Target	Fixed	Best fixed at test	Oracle	kNN (uncertainty)
PSNR 34 dB	50	30 (95–96%)	17	24.5 (100%)
PSNR 35 dB	100	50 (95%)	29	68 (100%)

S4b. Read noise and photon units

At $P = 30$, adding Gaussian read noise with standard deviation 0.005 or 0.01 (1.3 or 2.6 8-bit counts) changed the strong mixed-dose model from 36.70 to 36.68 and 36.66 dB, and fair TV deconvolution from 35.34 to 35.32 and 35.30 dB (`results/read_noise_fair.json`). The relative ranking is unchanged. The dose P is the Poisson scale at full 8-bit intensity; the mean detected photons per pixel on the test patches are 0.36, 3.56, and 35.6 at $P = 3, 30, 300$ (Poisson means; `results/photon_units.json`). The absolute photon counts should therefore be read as full-scale units, not as calibrated photons per emitter.

S4c. Doses not used in training

To test whether the model memorized the five training doses, we evaluated it on the 20 test FOVs at $P = 5, 50, 200$ (full-scale dose), which were never used for training. The measured PSNR is within 0.02 dB of log-dose interpolation between the neighboring trained doses (Table 8; `results/heldout_doses.json`).

Table 8: Strong mixed-dose model at held-out doses (PSNR, dB). Interpolation is linear in $\log_{10} P$ between the two neighboring training doses.

P	Neighbors	Interpolated	Measured	Residual
5	3, 10	33.32	33.32	0.00
50	30, 100	37.51	37.50	−0.01
200	100, 300	39.49	39.47	−0.02

S4d. Physical widefield PSF

The main simulation uses a Gaussian PSF. To test a physically motivated PSF, we repeated the $P = 3/30/300$ benchmark with an in-focus scalar Airy PSF for a circular aperture (NA 1.4, $\lambda = 520$ nm, 100 nm pixels). TV deconvolution was retuned on five validation FOVs with the correct Airy PSF. The Gaussian-trained strong model loses its advantage at high dose, while a model trained on a 50/50 Gaussian/Airy mixture (same 64-channel/48k/150 schedule) retains positive gains at all three doses (Table 9). On the Gaussian benchmark the same mixed model reaches 32.26/36.60/39.83 dB at $P = 3/30/300$, a loss of 0.01–0.18 dB relative to the Gaussian-trained model (`results/psf_airy.json`, `results/psf_airy_mix.json`, `results/psf_airy_mix_stats.json`, and `results/blind_psfmix_gauss.json`).

Table 9: Physical Airy PSF check (PSNR, dB). TV deconvolution uses the correct Airy PSF and validation-tuned parameters. "Gaussian-trained" is the main model; "PSF-mixed" is trained on a 50/50 Gaussian/Airy mixture.

P	Gaussian-trained	PSF-mixed	TV (Airy)	PSF-mixed gain
3	32.63	32.71	31.21	+1.50 [1.35, 1.65]
30	36.64	37.61	36.23	+1.39 [1.28, 1.48]
300	37.05	41.55	40.26	+1.29 [1.13, 1.45]

S5. Reproducibility

Data. The four archives were downloaded from `data.broadinstitute.org/bbbc`. SHA-256 checksums:

```
BBBC014 223c3230c164c8295f4626f82e4bd9fe394d8e9339a4f82faf697fdd9b835341
BBBC015 8c7c2abc579ff34173b49ccd06a167e1ad682f35506aa3f6e333d0336e1bad23
BBBC016 866ea3e6229a0f6fa2c9a18852b3274f7617326ecfae15d40f7f5e1123df7dd1
BBBC018 e2996b78e16de2eff74db649c1fe5335edfd82ded061e7e6afeadf27caa09df1
BBBC039 images 6f30a5d4fe38c928ded972704f085975f8dc0d65d9aa366df00e5a9d449fddd7
BBBC039 masks  f9e6043d8ca56344a4886f96a700d804d6ee982f31e2b2cd3194af2a053c2710
AI4Life Nuclei c7f2f5477a205e6020b74fded45d8f34f5383c82d261f5f401c1af58306dd7dc
AI4Life FMD    5110e958bef215e42649c2477855ea11f402842e11523ba0aebf05a634d7eb8f
```

Training and evaluation. The main mixed-dose model was trained with

```
python src/train_unet.py --dataset bbbc014 --n-val 20 --n-test 20 \
  --ch 64 --patches 48000 --epochs 150 --val-patches 4 --mixed-doses \
  --val-all-doses --out checkpoints/bbbc014_blind_strong.pt
```

The budget-matched strong conditioned variant adds `--dose-cond` and writes `bbbc014_dcond_strong.pt`; budget-matched specialists use `--photons 3`, 30, or 100 with the same 64-channel/48k/150 schedule (`bbbc014_spec*_strong.pt`). The PSF-mixed variant adds `--psf-mix` and writes `bbbc014_blind_psfmix.pt`. The base-budget ablation in Table 5 uses 24,000 patches and 100 epochs; its specialists use 16,000 patches and 60 epochs, and its phase-scrambled control uses `--dose-cond --scramble` with the 24,000-patch/100-epoch schedule. The budget-matched blind control uses the same 64-channel/48,000-patch/150-epoch mixed-dose schedule with `--scramble`. Evaluation uses `src/eval_blind_strong.py`, `src/eval_dcond_strong.py`, `src/eval_scr_strong.py`, `src/eval_spec_strong.py`, `src/eval_matched_ablation.py`, `src/bench_tvdeconv.py`, `src/eval_psf_airy.py`, and `src/bench_clean.py`. Figures and tables are regenerated with `src/make_fair_assets.py`, `src/make_fig1.py`, `src/make_fig2.py`, `src/make_fig3.py`, and `src/make_fig_real.py`.

Hardware. Training and inference used an NVIDIA A100-PCIE-40GB GPU. TV deconvolution and the CPU latency measurement used one core of the host CPU. The main split seed is 0; the test noise seed for the matched benchmark is 2026.

Patch-draw sensitivity. The headline values use the declared ten-patch, seed-2026 benchmark. To check the dependence on that draw, we repeated the comparison over five independent draws of 4 or 10 patches per FOV. The paired gain remained 0.80–1.01 dB at $P = 3$, 1.30–1.45 dB at $P = 30$, and 1.29–1.43 dB at $P = 300$; the absolute PSNR changes with the draw, but the gain does not (`results/patch_draw.json`). The GPU speed factor is computed from the rounded 1.2 ms ($103/1.2 = 86$); the unrounded 1.16 ms gives 89.

Code. The project repository contains the forward model, data loaders, splits, baselines, training scripts, evaluation scripts, result JSON files, checkpoints (with SHA-256 sums in `checkpoints/SHA256SUMS`), and figure scripts; the training commands above regenerate any omitted weight. No proprietary data are used.

S6. Downstream segmentation check

BBBC039 provides 200 Hoechst-stained FOVs with foreground masks and an official 100/50/50 split. The restoration model was trained on 12,000 patches for 40 epochs with doses including $P = 1$; it has no segmentation loss. TV deconvolution and TV denoising parameters were tuned per dose on the validation FOVs (not test), and Otsu [5] was applied identically to every method. Table 10 reports all 50 official test FOVs. With tuned TV, the restoration model improves Dice by only about 0.02 against the best tuned TV (0.795 versus 0.773 at $P = 1$, a gain of 0.022; 0.844 versus 0.824 at $P = 3$, a gain of 0.020). This downstream check therefore supports only a small utility gain and is not used as a main claim.

Table 10: BBBC039 nuclei Dice at selected doses (full-scale P); 50 official test FOVs, TV tuned per dose on validation, Otsu applied to every method.

Method	$P = 1$	$P = 3$	$P = 10$	$P = 30$	$P = 100$
Noisy + Otsu	0.192	0.344	0.434	0.575	0.769
TV denoise + Otsu	0.772	0.824	0.857	0.873	0.882
TV deconv + Otsu	0.773	0.815	0.843	0.870	0.885
U-Net + Otsu	0.795	0.844	0.878	0.890	0.898

S7. Real noisy/clean pairs

The real-data check uses the AI4Life-MDC25-Nuclei release (Zenodo 15624741, CC BY 4.0). The network has the same architecture as the main model with 48 channels in the first level and is trained on 128×128 patches from the 1,951 training pairs (16,000 patches, 80 epochs, batch 64, Adam, learning rate 10^{-3}, cosine schedule, flips and 90° rotations). Each image is normalized independently to its 0.5–99.9 percentile range. The checkpoint is selected on the 243 validation pairs (28.6 dB). Classical parameters are tuned on five validation pairs; Table 11 lists the grids and selected values (grid means archived in `results/real_small_tuning_grid.json`). Test pairs are never used for tuning or early stopping.

Table 11: Classical parameter tuning on five real validation pairs (mean PSNR in dB).

Method	Grid (PSNR, dB)	Selected
BM3D σ	0.2:22.60, 0.3:23.30, 0.4:23.30, 0.5:23.28, 0.7:23.24, 1.0:23.20	0.3
NLM h	0.1:22.06, 0.15:23.30, 0.2:23.35, 0.25:23.23, 0.3:23.11	0.2
TV w	0.08:21.44, 0.12:22.12, 0.16:22.14, 0.20:22.33, 0.24:22.45, 0.32:22.45, 0.4:22.38	0.24

Table 12 reports all 243 held-out test pairs. The network reaches 29.01 dB and 0.877 SSIM, versus 24.83 dB and 0.559 for affine-calibrated BM3D [6], 24.78 dB and 0.551 for NLM, 24.51 dB and 0.526 for TV, and 16.44 dB and 0.107 for the noisy input. The paired gain over BM3D is 4.18 dB (95% CI 3.94–4.42; exact Wilcoxon $p = 2.2e-41$; 240/243 wins), over NLM 4.23 dB, and over TV 4.50 dB. A second supervised baseline, a 17-layer DnCNN trained with the same 16,000 patches and 80 epochs, reaches 28.80 dB and 0.865

SSIM, 3.97 dB above affine-calibrated BM3D and 0.21 dB below the U-Net (95% CI 0.15–0.28; 210/243 wins). No pair is empty (minimum ground-truth standard deviation 0.047).

Affine calibration. The supervised networks are trained to predict the clean reference, so they can learn a global gain and offset that BM3D, NLM, and TV cannot. For the main comparison, each classical method receives one global affine map fitted on the five validation pairs and applied to the held-out outputs. The raw uncalibrated values are 21.30, 21.28, and 20.58 dB, giving a raw gain of about 7.7 dB; the affine-calibrated values are 24.83, 24.78, and 24.51 dB (gain 4.18 dB, 95% CI 3.94–4.42). Both raw and calibrated values are reported; the calibrated values are used in the main comparison. Refitting the maps on 5, 20, 50, or all 243 validation pairs changed the BM3D gain coefficient from 1.26 to 1.33 (NLM 1.30–1.40; TV 1.30–1.39). On a fixed 20-pair test subset, the resulting gain changed from 3.74 to 3.98 dB (`results/real_affine_stability.json` and `results/real_affine_243cal_20.json`), so the five-pair choice does not drive the reported effect.

Table 12: Real noisy/clean pairs from the AI4Life-MDC25 Jacquemet source. 243 held-out 256×256 pairs; classical parameters tuned on validation pairs. PSNR in dB.

Method	PSNR (dB)	SSIM
AI4Life nuclei, 243 held-out fields		
Noisy input	16.44	0.107
TV denoising	24.51	0.526
NLM	24.78	0.551
BM3D	24.83	0.559
DnCNN (supervised)	28.80	0.865
U-Net	29.01	0.877

Cross-source check. Applying the same weights to the 20 large BBBC006 fields gives 15.83 dB, below BM3D (17.05 dB), NLM (16.89 dB), and TV (16.45 dB); the noisy input is 13.35 dB. These fields have 16-bit saturated references and a different noise model. The failure is a domain shift, not a contradiction of the in-source result; it sets the scope of the real-data claim to matched acquisition.

A label-free variant trained only on the noisy images (J-invariant block mask, no clean targets) reached 17.89 dB and 0.196 SSIM on the same 243 pairs, 1.45 dB above the noisy input but 3.41 dB below tuned BM3D; a checkerboard mask learned the identity because the real camera noise is spatially correlated (`results/real_selfsup_all.json`). The paired clean references are therefore necessary for the present gain.

Reproduction. The pipeline is `src/train_real.py`, `src/tune_real_classical.py`, `src/eval_real_small_par.py`, `src/eval_real_large_par.py`, and `src/make_real_assets.py`; `src/make_fig_real.py` regenerates Fig. 4.

S7b. Second real source (FMD)

The FMD model uses the original base budget (32 channels, 8,000 patches, 40 epochs), not the stronger 48-channel nuclei model, and is trained on the 1,396 field-split FMD training pairs (301 validation, 301 held-out; one frame per field, and no field crosses splits; `datasets/fmd/split.json`). Classical baselines are tuned separately for each microscope on five validation fields. The selected BM3D sigmas are 0.12

(CF), 0.15 (TP), and 8.0 (WF); NLM uses $h = 0.1$ for all three; TV weights are 0.08 (CF) and 0.12 (TP, WF) (`results/fmd_tuning_micro.json`). On a pre-specified stratified subset of 150 of the 301 held-out fields (four per microscope/channel/level stratum, plus six fifth fields; the 150 fields are listed in `prereg/fmd_subset150.json`), the U-Net reaches 31.74 dB and 0.845 SSIM, versus 28.07 dB and 0.704 for BM3D, 26.96 dB and 0.697 for NLM, 27.01 dB and 0.707 for TV, and 25.62 dB for the noisy input. The paired gain over BM3D is 3.67 dB (95% CI 3.27–4.12; 147/150 fields, Wilcoxon $p = 2.9e-26$; `results/fmd_test150_micro.json` and `results/fmd_stats.json`). The gain is positive in all 36 strata, with group means from 0.4 to 11.1 dB; per microscope, the mean gains are +1.87 (CF), +3.41 (TP), and +5.96 (WF), and the largest gains are in WF. Baselines were tuned per microscope on five validation fields; BM3D grids were 0.05–0.3 for CF/TP and 3–20 for WF, and the selected sigmas 0.12, 0.15, and 8.0 are interior to those grids. The TV grid ended at 0.12 for TP and WF, and the selected weights lie at that boundary; we did not extend the grid, so the secondary TV comparison may slightly understate the tuned TV baseline. The FMD headline comparison is against BM3D. The FMD classical outputs are not affine-calibrated on FMD validation fields, so these numbers are not directly comparable to the nuclei table and should be read as a secondary upper bound. The remaining 151 test fields were not evaluated with BM3D because of its cost; the 150-field subset was pre-specified before prediction, and this is a secondary check. FMD is a separate underlying dataset in the same AI4Life release and required retraining, so it is not a zero-shot cross-source transfer.

Reproduction. The FMD pipeline in `src/` consists of `fmd_prepare.py`, `fmd_resplit.py`, `train_fmd.py`, `make_fmd_subset.py`, `tune_fmd_micro.py`, `eval_fmd_par.py`, `analyze_fmd150.py`, and `make_fmd_assets.py`.

S8. Task and dose identifiability probes

To check whether the acquisition parameters are available to the blind network, we trained probes to predict the dose (five levels on BBBC014) and the source task (BBBC014/015/016/018) from a single noisy 128×128 patch. Each probe uses 10 training and 10 test FOVs per class, 20 patches per FOV, with FOV-level splits. We compare a logistic regression on 14 simple features (intensity moments, percentiles, gradient energy, Laplacian variance, and three radial FFT bands) with a small CNN. The dose is identified with 97.9% (linear) and 100% (CNN) accuracy ($n = 1000/1000$; chance 20%), and the task with 79.5% and 75.8% accuracy ($n = 800/800$; chance 25%). The dose result supports the interpretation that the mixed-dose network can infer the noise level internally; the task result is consistent with the positive but smaller cross-dataset transfer. Code: `src/probe_task_dose.py`; result: `results/probe_task_dose.json`.

References

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